

Problem

Let M_{22} be the set of all 2×2 matrices. Is M_{22} a vector space? yes, see below for justification.

Let $A, B, C \in M_{22}$, and let a and $b \in \mathbb{R}$.

I) $A + B \in M_{22}$

(II), (III), (IV), and (V) follow from properties of matrix addition.

(VI) $aA \in M_{22}$

(VII), (VIII), (IX), and (X) follow from properties of scalar multiplication for matrices.

Definition

A **subspace** of a vector space V is a subset that is itself a vector space under the operations defined on V .

Problem

Let $\mathbb{M}_{22}$ be the set of all 2×2 matrices. Previously, we showed that $\mathbb{M}_{22}$ is a vector space. Is the following subset of $\mathbb{M}_{22}$ a subspace?

$$\mathbb{M}_L = \left\{ \begin{bmatrix} x & 0 \\ y & z \end{bmatrix} \mid x + y + z = 0 \right\}$$

Let $A = \begin{bmatrix} a_1 & 0 \\ a_2 & a_3 \end{bmatrix}$, $B = \begin{bmatrix} b_1 & 0 \\ b_2 & b_3 \end{bmatrix}$, and $C = \begin{bmatrix} c_1 & 0 \\ c_2 & c_3 \end{bmatrix}$ be in $\mathbb{M}_L$, then:

(I) $A + B = \begin{bmatrix} a_1 & 0 \\ a_2 & a_3 \end{bmatrix} + \begin{bmatrix} b_1 & 0 \\ b_2 & b_3 \end{bmatrix} = \begin{bmatrix} a_1 + b_1 & 0 \\ a_2 + b_2 & a_3 + b_3 \end{bmatrix}$ is also in $\mathbb{M}_L$.

(II) $A + B = B + A$

(III) $0_{2 \times 2} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \in \mathbb{M}_L$ and $A + 0 = A$.

(IV) $(A + B) + C = A + (B + C)$ follows from properties of matrix addition.

(V) $-A = \begin{bmatrix} -a_1 & 0 \\ -a_2 & -a_3 \end{bmatrix} \in \mathbb{M}_L$ and $A + (-A) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

(VI) $A = \begin{bmatrix} a_1 & 0 \\ a_2 & a_3 \end{bmatrix} \in \mathbb{M}_L$, hence $a_1 + a_2 + a_3 = 0$. Then, $aA = \begin{bmatrix} aa_1 & 0 \\ aa_2 & aa_3 \end{bmatrix} \in \mathbb{M}_L$, since $aa_1 + aa_2 + aa_3 = 0$.

Similarly, you can check that (VII), (VIII), (IX), and (X) also hold.

Question: Can you modify $\mathbb{M}_L$ such that the new subset of $\mathbb{M}_{2 \times 2}$ is no longer a subspace?

Theorem

In a vector space, the span of any subset is a subspace.

Example

Consider the vector space of quadratic polynomials $\mathbb{P}_2$.

- Show that $\{1, x, x^2\}$ is a basis for $\mathbb{P}_2$. ($\{1, x, x^2\}$ is known as the standard basis for $\mathbb{P}_2$.)
- Is the span of the set $\{x - x^2, 3x^2\}$ a subspace of $\mathbb{P}_2$? If so, is $\{x - x^2, 3x^2\}$ a basis for this subspace?

● Note that any quadratic polynomial is a linear combination of $1, x$, and x^2 :

$$P(x) = a_2 x^2 + a_1 x + a_0(1).$$

Also, clearly none of the elements of the set $\{1, x, x^2\}$ can be written as a linear combination of the rest of the elements. Hence the set $\{1, x, x^2\}$ is linearly independent, which means it is a basis for $\mathbb{P}_2$.

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- Yes, the set $\{x - x^2, 3x^2\}$ is a subset of $\mathbb{P}_2$ and hence its span is a subspace of $\mathbb{P}_2$.
Also, the set $\{x - x^2, 3x^2\}$ is a linearly independent set, and hence it is a basis for this subspace.

Reminders

Definition

Let H be a nonempty collection of vectors in $\mathbb{R}^n$. Then H is a **subspace** if whenever a and b are scalars and $\vec{u}$ and $\vec{v}$ are vectors in H , $a\vec{u} + b\vec{v}$ is also in H .

Theorem

Let H be a nonempty collection of vectors in $\mathbb{R}^n$. Then H is a subspace of $\mathbb{R}^n$ if and only if there exist vectors $\{\vec{u}_1, \dots, \vec{u}_k\}$ in H such that

$$H = \text{span} \{ \vec{u}_1, \dots, \vec{u}_k \}$$

Definition

Let H be a subspace of $\mathbb{R}^n$. Then $\{\vec{u}_1, \dots, \vec{u}_k\}$ is called a **basis** for H if the following conditions hold:

- 1 $\text{span}\{\vec{u}_1, \dots, \vec{u}_k\} = H$
- 2 $\{\vec{u}_1, \dots, \vec{u}_k\}$ is linearly independent.

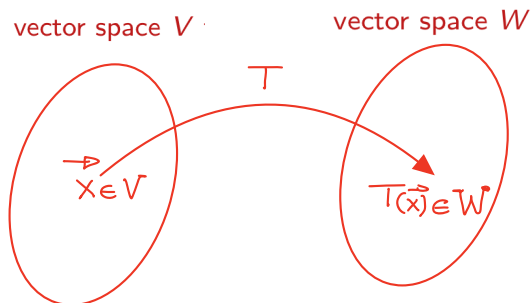
Definition

A linear transformation from a vector space V to another vector space W is a rule that assigns to each $\vec{x} \in V$ a unique vector $T(\vec{x}) \in W$ and it preserves vector addition and scalar multiplication.

For every $\vec{x}_1$ and $\vec{x}_2 \in V$, and $a_1, a_2 \in \mathbb{R}$, we have:

$$T(a_1\vec{x}_1 + a_2\vec{x}_2) = a_1T(\vec{x}_1) + a_2T(\vec{x}_2)$$

T preserves linear combinations.



Remark:

The null space and the column space of an $m \times n$ matrix are subspaces of $\mathbb{R}^n$ and $\mathbb{R}^m$ respectively.

Here we show that null space of an $m \times n$ matrix is a subspace of $\mathbb{R}^n$ and we leave the column space as an exercise.

$$\text{Nul } A = \text{Ker } A = \{ \vec{x} \in \mathbb{R}^n \mid A_{m \times n} \vec{x}_{n \times 1} = \vec{0}_{m \times 1} \}$$

Assume $a, b \in \mathbb{R}$ and $\vec{x}_1, \vec{x}_2 \in \text{Nul } A$, then

$$A(a\vec{x}_1 + b\vec{x}_2) = aA(\vec{x}_1) + bA(\vec{x}_2) = \vec{0} + \vec{0} = \vec{0} \iff a\vec{x}_1 + b\vec{x}_2 \in \text{Nul } A$$

Hence $\text{Nul } A$ is a subspace of $\mathbb{R}^n$.

A linear map T is called an **isomorphism** if the following two conditions are satisfied.

- T is one to one. That is, if $T(\vec{x}) = T(\vec{y})$, then $\vec{x} = \vec{y}$.
- T is onto. That is, if $\vec{w} \in W$, there exists $\vec{v} \in V$ such that $T(\vec{v}) = \vec{w}$.

Two such subspaces which have an isomorphism as described above are said to be **isomorphic**.

Remark: We consider isomorphic spaces to be the same, because there is a one-to-one correspondence between their elements and the structure of the vector space operations is preserved through linearity.

Example: Show that $\mathbb{R}^n$ and P_{n-1} are isomorphic.

We can find an isomorphism between these two spaces. Consider $T: \mathbb{R}^n \rightarrow P_{n-1}$ defined by $T\left(\begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_{n-1} \end{bmatrix}\right) = a_0 + a_1x + \dots + a_{n-1}x^{n-1}$.

Note that T is linear, one-to-one, and onto.

Example: Consider the transformation $\mathbb{P}_2 \xrightarrow{T} \mathbb{R}^2$ defined as

$$T(P(x)) = \begin{bmatrix} P(0) \\ P'(1) \end{bmatrix}.$$

① Using the standard basis for $\mathbb{P}_2$ and $\mathbb{R}^2$, find the matrix of T .

$$P(x) = ax^2 + bx + c \quad P'(x) = 2ax + b$$

$$T(P(x)) = \begin{bmatrix} P(0) \\ P'(1) \end{bmatrix} = \begin{bmatrix} c \\ 2a + b \end{bmatrix}$$

Diagram illustrating the transformation T and its matrix representation. The diagram shows $\mathbb{P}_2$ and $\mathbb{R}^3$ connected by a green double-headed arrow labeled "Isomorphism". A red arrow labeled T points from $\mathbb{P}_2$ to $\mathbb{R}^2$.

$$A = \begin{bmatrix} T(\vec{e}_1) & T(\vec{e}_2) & T(\vec{e}_3) \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \end{bmatrix}$$
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} c \\ b \\ a \end{bmatrix} = \begin{bmatrix} c \\ 2a + b \end{bmatrix}$$

⑤ Is T one-to-one? No, since $A\vec{x} = \vec{0}$ has non-trivial solutions:

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 0 \end{array} \right] \rightarrow \vec{x} = \begin{bmatrix} c \\ b \\ a \end{bmatrix} = \begin{bmatrix} 0 \\ -2t \\ t \end{bmatrix} = t \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}$$

$$\left. \begin{array}{l} T(x^2 - 2x) = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ T(-x^2 + 2x) = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \end{array} \right\} \Rightarrow T \text{ is NOT one-to-one.}$$